Precautions

1. When measuring DC voltage, care must be taken while connecting the two probes of the meter with the DC source. It should be connected with correct polarities in parallel connection with DC supply. That is, positive probe of the meter should be connected to positive terminal of the DC supply and negative probe to negative terminal of the DC supply as shown in Figure 10.11(a).

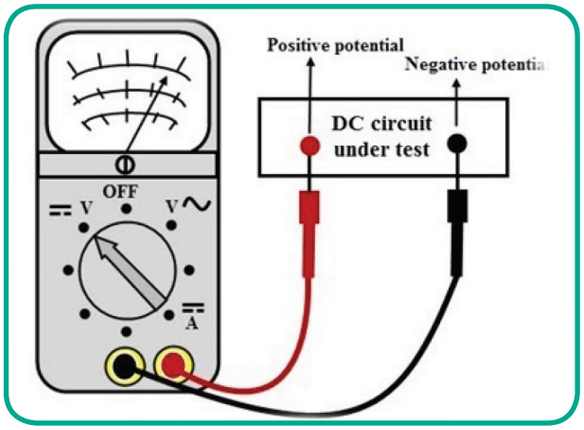


Figure 10.11 (a) Connecting a Multimeter while Measuring DC Voltage

2. When measuring DC current, multimeter should be connected in series with the DC supply and the load with correct polarities. That is, positive probe should be connected with positive terminal of the supply and negative probe to the load as shown in Figure 10.11(b).

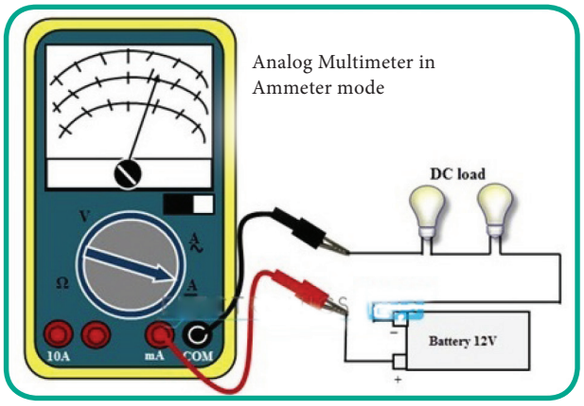


Figure 10.11 (b) Connecting a Multimeter while Measuring DC Current

3. In the above two measurements, if we wrongly connect the probes, the meter needle will get reversed and subsequently the meter will get damaged.
4. While measuring AC voltages, it should be connected in parallel but polarity need not be considered as shown in Figure 10.12. Because AC supply has no fixed polarities (+ and -).

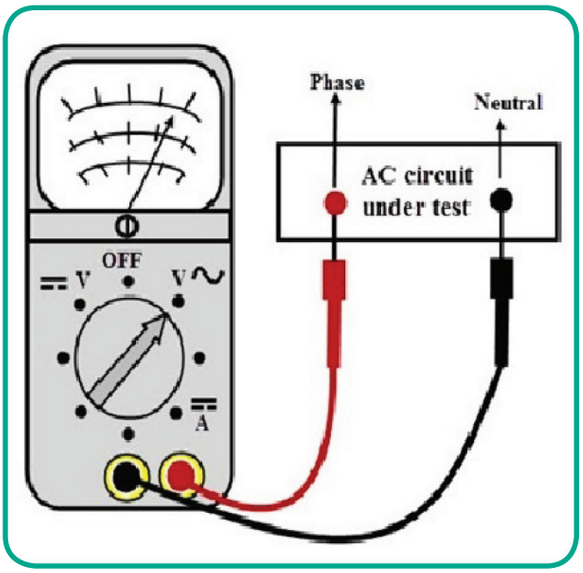


Figure 10.12 Connecting a multimeter while measuring AC voltage

5. While measuring AC current, as like the measurement of DC current, the meter should be connected in series with the supply and the load as shown in Figure 10.13. Polarity need not be considered as like in measuring AC voltage.

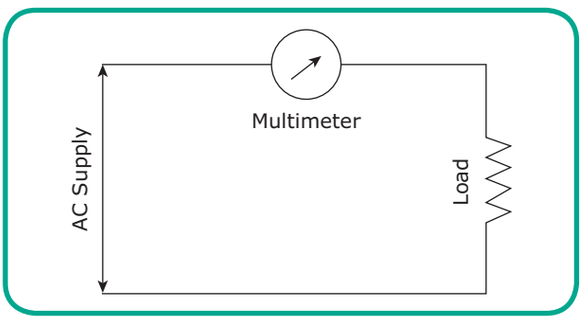


Figure 10.13 AC current Measurement

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